

Partial Fractions

Section 8.5

2-2-26

A) Distinct Linear Factors

$$\int \frac{1}{x^2 - x - 6} dx =$$

$$\frac{1}{(x-3)(x+2)} = \frac{A}{(x-3)} + \frac{B}{(x+2)}$$

$$1 = A(x+2) + B(x-3)$$

$$x = -2 \quad 1 = -5B \quad B = -\frac{1}{5}$$

$$x = 3 \quad 1 = 5A \quad A = \frac{1}{5}$$

$$\frac{1}{5} \int \frac{1}{x-3} - \frac{1}{x+2} dx$$

$$\frac{1}{5} \int \frac{1}{x-3} - \frac{1}{5} \int \frac{1}{x+2}$$

$$\boxed{\frac{1}{5} (\ln|x-3| - \ln|x+2|) + C}$$

$$\text{Ex: } \int \frac{x+8}{x^2+6x+8} dx =$$

$$\int \frac{x+8}{(x+4)(x+2)} dx$$

$$\frac{A}{(x+4)} + \frac{B}{(x+2)}$$

$$x+8 = A(x+2) + B(x+4)$$

$$x = -2: 6 = 2B \quad B = 3$$

$$x = -4: 4 = -2A \quad A = -2$$

$$\int \frac{3}{x+2} - \frac{2}{x+4} dx =$$

$$= \boxed{3 \ln|x+2| - 2 \ln|x+4| + C}$$

B) Repeating Linear Factors

$$\text{Ex: } \int \frac{3x}{x^2-4x+4} dx$$

$$\frac{A}{(x-2)} + \frac{B}{(x-2)^2}$$

$$3x = A(x-2) + B$$

$$x=2: B=6$$

$$x=R: A=3$$

$$\int \frac{3}{x-2} + \frac{6}{(x-2)^2} dx$$

$$\boxed{3 \ln|x-2| - \frac{6}{x-2} + C}$$

C) Quadratic Factor

$$\text{Ex: } \int \frac{5x^2-x+27}{x^3+9x} dx$$

$$5x^2-x+27 = \frac{A}{x} + \frac{Bx+C}{(x^2+9)}$$

$$5x^2-x+27 = A(x^2+9) + x(Bx+C)$$

$$Ax^2+9A+Bx^2+Cx$$

$$x=0: 27=9A \quad A=3$$

$$C=-1$$

$$5=A+B \quad B=2$$

$$A=3, B=2, C=-1$$

$$\int \frac{3}{x} + \frac{2x-1}{x^2+9} dx$$

$$\int \frac{3}{x} + \int \frac{2x}{x^2+9} - \int \frac{1}{x^2+9} dx$$

$$\boxed{3 \ln|x| + \ln|x^2+9| - \frac{1}{3} \arctan\left(\frac{x}{3}\right) + C}$$